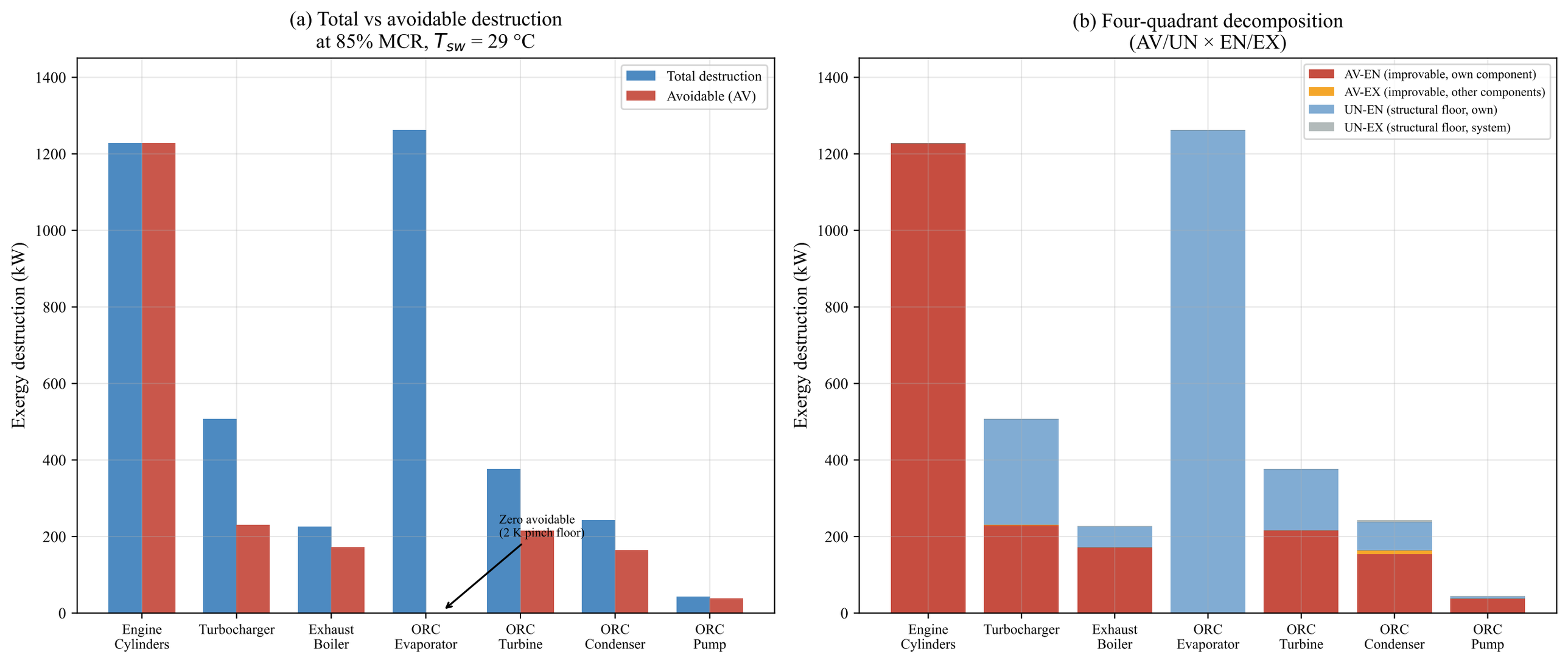

(a) Total vs avoidable destruction
at 85% MCR, T_{sw} = 29 °C
Exergy destruction (kW)
0
200
400
600
800
1000
1200
1400
Total destruction
Avoidable (AV)
Zero avoidable
(2 K pinch floor)
Engine Cylinders
Turbocharger
Exhaust Boiler
ORC Evaporator
ORC Turbine
ORC Condenser
ORC Pump
(b) Four-quadrant decomposition
(AV/UN × EN/EX)
Exergy destruction (kW)
AV-EN (improvable, own component)
AV-EX (improvable, other components)
UN-EN (structural floor, own)
UN-EX (structural floor, system)
Engine Cylinders
Turbocharger
Exhaust Boiler
ORC Evaporator
ORC Turbine
ORC Condenser
ORC Pump